

ML2025 Fall Homework Assignment 5

Handwritten

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Tools You Need to Know/Learn

- Markov decision process, Q value function and value function
- Stationary and Deterministic Policy
- Policy iteration, Performance difference
- Projection matrix and symmetric matrix.
- State visitation distribution, Bayesian inference

You are expected to know (or learn through this homework) the definitions and usages of the above concepts, and are encouraged to discuss with TAs during TA hour if you encounter difficulties.

Homework Policy

- The homework is graded out of 100 points, with up to 15 additional bonus points available (only awarded for completely correct solutions).
- The official submission deadline will follow the schedule announced on NTU COOL.
- Homework may be handwritten or typed (e.g. using $\text{\LaTeX}$), but must be submitted in **PDF format**.
- If you discuss the homework with classmates, you should state their student IDs in your submission.
- Plagiarism is strictly prohibited. Serious violations will be dealt with according to NTU regulations.

Problem 1 (Trace Optimization)

1. Let $\Sigma \in \mathbb{R}^{m \times m}$ be a symmetric positive semi-definite matrix, $\mu \in \mathbb{R}^m$. Please construct a set of points $x_1, \dots, x_n \in \mathbb{R}^m$ such that

$$\frac{1}{n} \sum_{i=1}^n (x_i - \mu)(x_i - \mu)^T = \Sigma, \quad \frac{1}{n} \sum_{i=1}^n x_i = \mu$$

Hint: n is not given by the problem. WLOG, you may assume $\mu = 0 \in \mathbb{R}^d$.

Solution: WLOG (Without Loss of Generality), let $\mu = 0$. Since Σ is a symmetric positive semi-definite matrix, we can perform eigen decomposition as follows:

$$\Sigma = UDU^T = \sum_{i=1}^m (d_i u_i u_i^T).$$

where U and U^T are orthogonal matrix. Let $n = 2m$ and construct a set of points $x_1, \dots, x_m, \dots, x_{2m}$ where $x_i = \sqrt{md_i} u_i$ and $x_{m+i} = -\sqrt{md_i} u_i \forall 1 \leq i \leq m$. Then,

$$\begin{aligned} \frac{1}{n} \sum_{i=1}^n x_i &= \mu = 0 \\ \frac{1}{n} \sum_{i=1}^n x_i x_i^T &= \sum_{i=1}^m (d_i u_i u_i^T) = UDU^T = \Sigma \end{aligned}$$

Note that lots of students just use the covariance of eigen decomposition to construct $\{x_i\}_{i=1}^n$. However, It should satisfy the condition that $\frac{1}{n} \sum_{i=1}^n x_i = \mu$

2. Let $1 \leq k \leq m$, solve the following optimization problem and justify with proof:

minimize $\text{Trace}(\Phi^T \Sigma \Phi)$
 subject to $\Phi^T \Phi = I_k$
 variables $\Phi \in \mathbb{R}^{m \times k}$

In other words, you need to find Φ and verify that your Φ minimize the trace.

Solution: Method 1: Let $\phi_1, \dots, \phi_k$ be the columns of Φ . Then

$$\text{Trace}(\Phi^T \Sigma \Phi) = \sum_{i=1}^k \phi_i^T \Sigma \phi_i = \sum_{i=1}^k \phi_i^T \left(\sum_{j=1}^m (d_j u_j u_j^T) \right) \phi_i = \sum_{j=1}^m d_j \sum_{i=1}^k \langle u_j, \phi_i \rangle^2 = \sum_{j=1}^m c_j d_j$$

where $\langle \cdot, \cdot \rangle$ is standard inner product in Euclidean space, $c_j := \sum_{i=1}^k \langle u_j, \phi_i \rangle^2$ for each $j = 1, \dots, m$ and $d_1 \geq d_2 \geq \dots \geq d_m \geq 0$ Claim: $0 \leq c_j \leq 1$ and $\sum_{j=1}^m c_j = k$ Clearly, $c_j \geq 0$. Extending $\phi_1, \dots, \phi_k$ to $\phi_1, \dots, \phi_k, \phi_{k+1}, \dots, \phi_m$ for $\mathbb{R}^m$. Then, for each $j = 1, \dots, m$

$$c_j = \sum_{i=1}^k \langle u_j, \phi_i \rangle^2 \leq \sum_{i=1}^m \langle u_j, \phi_i \rangle^2 = 1$$

Finally, since $u_1, \dots, u_m$ is an orthonormal basis for $\mathbb{R}^m$,

$$\sum_{j=1}^m c_j = \sum_{j=1}^m \sum_{i=1}^k \langle u_j, \phi_i \rangle^2 = \sum_{i=1}^k \sum_{j=1}^m \langle u_j, \phi_i \rangle^2 = \sum_{i=1}^k \|\phi_i\|_2^2 = k$$

Hence, the minimum value of $\sum_{j=1}^m c_j d_j$ over all choice of $c_1, c_2, \dots, c_m \in [0, 1]$ with $\sum_{j=1}^m c_j = k$ is $d_{m-k+1}, \dots, d_m$. This is achieved when $c_1, \dots, c_{m-k} = 0$ and $c_{m-k+1} = \dots = c_m = 1$.

Notice: Many students proceed the proof as the following. By the symmetry of Σ , we have

$$\Sigma = U \Lambda U^T, \quad U \text{ orthonormal.}$$

Let $V = U^T \Phi$ and hence $\Phi = UV$. Then

$$\text{Tr}(\Phi^T \Sigma \Phi) = \text{Tr}(\Phi^T U \Lambda U^T \Phi) = \text{Tr}(V^T \Lambda V).$$

Note that $\Lambda \in \mathbb{R}^{m \times m}$, $V \in \mathbb{R}^{m \times k}$, and $V^T \Lambda V \in \mathbb{R}^{k \times k}$. Then

$$\text{Tr}(V^T \Lambda V) = \sum_{i=1}^k \lambda_i v_i^T v_i = \sum_{i=1}^k \lambda_i$$

“because V is orthonormal”, which is not true because $U^T \Phi$ is not necessarily orthonormal.

Problem 2 (Gradient Boosting)

Consider the binary classification problem, where we are given training data set $\{(x_i, y_i)\}_{i=1}^N$ with $x_i \in \mathbb{R}^d$ and $y_i \in \{1, -1\}$. Let $F = \{f \mid f : \mathbb{R}^d \rightarrow \{1, -1\}\}$ be the collection of classifiers. Given number of epochs $T \in \mathbb{N}$. Suppose that we want to find the function

$$g_{T+1}(x) = \sum_{t=1}^T \alpha_t f_t(x)$$

where $f_t \in F$ and $\alpha_t \in \mathbb{R}$ for all $t = 1, \dots, T$, by which the aggregated classifier is given by

$$h(x) = \begin{cases} 1, & \text{if } g(x) > 0 \\ -1, & \text{if } g(x) \leq 0. \end{cases}$$

Please apply gradient boosting to show how the functions f_t and the coefficients α_t are computed with an aim to minimize the following loss function

$$L(g) = \sum_{i=1}^N \log \left(1 + e^{-y_i g(x_i)} \right).$$

Solution:

Choose f_t Compute the derivative

$$\frac{\partial \ell_i}{\partial g(x_i)} = -\frac{y_i e^{-y_i g(x_i)}}{1 + e^{-y_i g(x_i)}} = -\frac{y_i}{1 + e^{y_i g(x_i)}}.$$

Hence,

$$r_{t,i} = -\left. \frac{\partial \ell_i}{\partial g(x_i)} \right|_{g=g_{t-1}} = \frac{y_i}{1 + e^{y_i g_{t-1}(x_i)}} = y_i w_{t,i},$$

where

$$w_{t,i} = \frac{1}{1 + e^{y_i g_{t-1}(x_i)}}.$$

Choose the weak learner via squared error

$$f_t = \arg \min_{f \in \mathcal{F}} \sum_{i=1}^N (r_{t,i} - f(x_i))^2.$$

Since $f(x_i) \in \{-1, +1\}$,

$$f_t = \arg \max_{f \in \mathcal{F}} \sum_{i=1}^N r_{t,i} f(x_i) = \arg \max_{f \in \mathcal{F}} \sum_{i=1}^N w_{t,i} y_i f(x_i).$$

Note that

$$y_i f(x_i) = \begin{cases} 1, & f(x_i) = y_i, \\ -1, & f(x_i) \neq y_i. \end{cases}$$

Therefore,

$$f_t = \arg \min_{f \in \mathcal{F}} \sum_{i: f(x_i) \neq y_i} w_{t,i}.$$

Choose α_t By minimizing the loss along f_t ,

$$\alpha_t \in \operatorname{argmin}_{\alpha \in \mathbb{R}} L(g_{t-1} + \alpha f_t) = \operatorname{argmin}_{\alpha \in \mathbb{R}} \sum_{i=1}^N \log(1 + e^{-y_i(g_{t-1}(x_i) + \alpha f_t(x_i))}).$$

Equivalently, we may characterize α_t as the solution of

$$\left. \frac{\partial}{\partial \alpha} L(g_{t-1} + \alpha f_t) \right|_{\alpha=\alpha_t} = 0,$$

i.e.

$$\sum_{i=1}^N \frac{-y_i f_t(x_i) e^{-y_i(g_{t-1}(x_i) + \alpha_t f_t(x_i))}}{1 + e^{y_i(g_{t-1}(x_i) + \alpha_t f_t(x_i))}} = \sum_{\{i | f_t(x_i) = y_i\}} \frac{-e^{-y_i g_{t-1}(x_i) - \alpha_t}}{1 + e^{-y_i g_{t-1}(x_i) - \alpha_t}} + \sum_{\{i | f_t(x_i) \neq y_i\}} \frac{-e^{-y_i g_{t-1}(x_i) + \alpha_t}}{1 + e^{-y_i g_{t-1}(x_i) + \alpha_t}} = 0$$

This one-dimensional optimization (or root-finding) in α can be solved numerically (line search or Newton method).

Problem 3 (EM algorithm for mixture of exponential model)

Given N samples $x_1, \dots, x_N \in [0, \infty)$, we would like to cluster them into K clusters. Assume the samples are generated according to Exponential mixture models

$$X \sim \sum_{j=1}^K \pi_j \text{Exp}(\tau_j)$$

where $\pi_1 + \dots + \pi_K = 1$, and $\text{Exp}(\tau)$ denotes the exponential distribution with probability density function

$$f_\tau(x) = \begin{cases} (1/\tau)e^{-x/\tau} & , x \geq 0 \\ 0 & , x < 0. \end{cases}$$

We would like to apply Expectation Maximization algorithm to find the maximum likelihood estimation of parameters $\theta = \{(\pi_k, \tau_k)\}_{k=1}^K$.

- (a) Please write down the E-step and M-step and show that the parameters are updated from $\theta^{(t)} = \{(\pi_k^{(t)}, \tau_k^{(t)})\}_{k=1}^K$ to $\theta^{(t+1)} = \{(\pi_k^{(t+1)}, \tau_k^{(t+1)})\}_{k=1}^K$ in the following form:

$$\tau_k^{(t+1)} = \frac{\sum_{i=1}^N \delta_{ik}^{(t)} x_i}{\sum_{i=1}^N \delta_{ik}^{(t)}}, \quad \pi_k^{(t+1)} = \frac{1}{N} \sum_{i=1}^N \delta_{ik}^{(t)}$$

- (b) What is the closed form expression of $\delta_{ik}^{(t)}$?

Solution:

Denote $\mathcal{X} = (\mathbf{x}_i)_{i=1}^N$ and latent variables $\mathcal{Z} = (\mathbf{z}_1, \dots, \mathbf{z}_N) \in \{1, \dots, K\}^N$, and consider the joint probability distribution as

$$p(\mathcal{X}, \mathcal{Z}; \theta) = \prod_{i=1}^N p(\mathbf{x}_i, z_i; \theta), \quad \text{where } p(\mathbf{x}, z = k; \theta) = \pi_k \cdot \frac{1}{\tau_k} e^{-x_i/\tau_k}, \quad x_i \geq 0.$$

Start with initial guess $\theta^{(0)}$, the EM algorithm estimates $\theta^{(t+1)} = (\pi_k^{(t+1)}, \mu_k^{(t+1)})_{k=1}^K$ from $\theta^{(t)} = (\pi_k^{(t)}, \mu_k^{(t)})_{k=1}^K$.

Expectation Step Compute

$$Q(\theta \mid \theta^{(t)}) = \mathbb{E}_{\mathcal{Z} \mid \mathcal{X}; \theta^{(t)}} [\log p(\mathcal{X}, \mathcal{Z}; \theta)] = \sum_{i=1}^N \mathbb{E}_{z_i \mid \mathbf{x}_i; \theta^{(t)}} [\log p(\mathbf{x}_i, z_i; \theta)].$$

The posterior probability of latent variables based on current parameters $\theta^{(t)}$

$$P[z_i = k \mid \mathbf{x}_i; \theta^{(t)}] = \frac{p(\mathbf{x}_i, z_i = k; \theta^{(t)})}{\sum_{k'=1}^K p(\mathbf{x}_i, z_i = k'; \theta^{(t)})} = \frac{\pi_k \cdot \frac{1}{\tau_k} e^{-x_i/\tau_k}}{\sum_{k'=1}^K \pi_{k'} \cdot \frac{1}{\tau_{k'}} e^{-x_i/\tau_{k'}}}.$$

Hence,

$$Q(\theta \mid \theta^{(t)}) = \sum_{i=1}^N \sum_{k=1}^K \delta_{ik}^{(t)} \left(\log \pi_k - \log \tau_k - \frac{x_i}{\tau_k} \right),$$

where $\delta_{ik}^{(t)} = P[z_i = k \mid \mathbf{x}_i; \theta^{(t)}]$.

Maximization Step Choose $\theta^{(t+1)} = \arg \max_{\theta} Q(\theta \mid \theta^{(t)})$. Note that

$$\frac{\partial}{\partial \tau_k} Q(\theta \mid \theta^{(t)}) = \sum_{i=1}^N \delta_{ik}^{(t)} \left(-\frac{1}{\tau_k} + \frac{x_i}{\tau_k^2} \right).$$

By setting the partial derivative to zero, the maximal solution for τ_k is

$$\tau_k^{(t+1)} = \frac{\sum_{i=1}^N \delta_{ik}^{(t)} x_i}{\sum_{i=1}^N \delta_{ik}^{(t)}}.$$

As for π_k , due to the constraint $\sum_{k=1}^K \pi_k = 1$, we introduce Lagrange multipliers and note

$$\frac{\partial}{\partial \pi_k} \left(Q(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(t)}) - \lambda \sum_{k=1}^K \pi_k \right) = \frac{1}{\pi_k} \sum_{i=1}^N \delta_{ik}^{(t)} - \lambda.$$

By setting the above quantities identically zero for all k , we solve $\lambda = N$ and the maximal solution for π_k is

$$\pi_k^{(t+1)} = \frac{1}{N} \sum_{i=1}^N \delta_{ik}^{(t)}.$$

Problem 4 (Sparse SVM)

Given training data of N input-output pairs $\mathcal{D} = ((x_i, y_i))_{i=1}^N$, where $x_i \in \mathcal{X}$ and $y_i \in \{\pm 1\}$. One can give two types of arguments in favor of the SVM algorithm: one based on the sparsity of the support vectors, another based on the notion of margin. Suppose instead of maximizing the margin, we choose instead to maximize sparsity by minimizing the p -norm of the vector $\alpha = (\alpha_1, \dots, \alpha_N)$ that defines the weight vector $\mathbf{w}$, for some $p \geq 1$. In this question we consider the case $p = 2$, which leads to the following optimization problem:

$$\begin{aligned} & \text{minimize} && f(\alpha, b, \xi) = \frac{1}{2} \sum_{i=1}^N \alpha_i^2 + \sum_{i=1}^N C_i \xi_i \\ & \text{subject to} && y_i \left(\sum_{j=1}^N \alpha_j y_j \mathbf{x}_i \cdot \mathbf{x}_j + b \right) \geq 1 - \xi_i, \quad i \in \{1, \dots, N\} \\ & \text{variables} && b \in \mathbb{R}, \alpha_i \geq 0, \xi_i \geq 0, \quad i \in \{1, \dots, N\} \end{aligned}$$

which can be rewritten in the following primal problem:

$$\begin{aligned} & \text{minimize} && f(\alpha, b, \xi) = \frac{1}{2} \sum_{i=1}^N \alpha_i^2 + \sum_{i=1}^N C_i \xi_i \\ & \text{subject to} && \left. \begin{aligned} g_{1,i}(\alpha, b, \xi) &= 1 - \xi_i - y_i \left(\sum_{j=1}^N \alpha_j y_j \mathbf{x}_i \cdot \mathbf{x}_j + b \right) \leq 0 \\ g_{2,i}(\alpha, b, \xi) &= -\alpha_i \leq 0 \\ g_{3,i}(\alpha, b, \xi) &= -\xi_i \leq 0 \end{aligned} \right\} \quad i \in \{1, \dots, N\} \\ & \text{variables} && \alpha = (\alpha_1, \dots, \alpha_N) \in \mathbb{R}^N, b \in \mathbb{R}, \xi = (\xi_1, \dots, \xi_N) \in \mathbb{R}^N \end{aligned} \quad (1)$$

as well as its Lagrangian dual problem:

$$\begin{aligned} & \text{maximize} && \theta(\omega, \beta, \gamma) = \inf_{\alpha \in \mathbb{R}^N, b \in \mathbb{R}, \xi \in \mathbb{R}^N} L(\alpha, b, \xi, \omega, \beta, \gamma) \\ & \text{subject to} && \omega_i \geq 0, \beta_i \geq 0, \gamma_i \geq 0, \quad i \in [1, N] \\ & \text{variables} && \omega = (\omega_1, \dots, \omega_N) \in \mathbb{R}^N, \beta = (\beta_1, \dots, \beta_N) \in \mathbb{R}^N, \gamma = (\gamma_1, \dots, \gamma_N) \in \mathbb{R}^N \end{aligned} \quad (2)$$

1. Write down the Lagrangian function $L(\alpha, b, \xi, \omega, \beta, \gamma)$ in explicit form of $\alpha, b, \xi, \omega, \beta, \gamma$.
2. Show that the duality gap between (1) and (2) is zero.
3. Derive $\theta(\omega, \beta, \gamma)$ in explicit form of dual variables ω, β, γ .
4. Show that the dual problem can be simplified as

$$\begin{aligned} & \text{maximize} && \sum_{i=1}^N \omega_i - \frac{1}{2} \sum_{i=1}^N \left(\sum_{j=1}^N \omega_j y_j y_i \mathbf{x}_j \cdot \mathbf{x}_i \right)_+^2 \\ & \text{subject to} && \sum_{i=1}^N \omega_i y_i = 0 \\ & \text{variables} && 0 \leq \omega_i \leq C_i, \quad i = 1, \dots, N \end{aligned} \quad (3)$$

5. Suppose $(\bar{\alpha}, \bar{b}, \bar{\xi})$ and $(\bar{\omega}, \bar{\beta}, \bar{\gamma})$ are the optimal solutions to problems (1) and (2) respectively. Denote $\bar{\mathbf{w}} = \sum_{j=1}^N \bar{\alpha}_j y_j \mathbf{x}_j$.

(a) Prove that

$$\bar{\alpha}_i = \max \left(\sum_{j=1}^N \bar{\omega}_j y_j y_i \mathbf{x}_j \cdot \mathbf{x}_i, 0 \right) \quad \forall i = 1, \dots, N \quad (4)$$

(b) Prove that

$$\bar{b} = \arg \min_{b \in \mathbb{R}} \sum_{i=1}^N C_i \max(1 - y_i (\bar{\mathbf{w}} \cdot \mathbf{x}_i + b), 0), \quad (5)$$

(c) Prove that $\bar{\xi}_i = \max(1 - y_i (\bar{\mathbf{w}} \cdot \mathbf{x}_i + \bar{b}), 0)$ for all $i = 1, \dots, N$.

(d) Prove that

$$\left. \begin{aligned} \bar{\omega}_i &= C_i, && \text{if } y_i (\bar{\mathbf{w}} \cdot \mathbf{x}_i + \bar{b}) < 1 \\ \bar{\omega}_i &= 0, && \text{if } y_i (\bar{\mathbf{w}} \cdot \mathbf{x}_i + \bar{b}) > 1 \\ 0 \leq \bar{\omega}_i &\leq C_i, && \text{if } y_i (\bar{\mathbf{w}} \cdot \mathbf{x}_i + \bar{b}) = 1 \end{aligned} \right\} \quad \forall i = 1, \dots, N$$

Solution:

1 The Lagrangian function can be explicitly written as

$$L(\alpha, \beta, \omega, \gamma) = f(\alpha, b, \xi) + \sum_{i=1}^N \omega_i g_1^i(\alpha, b, \xi) + \sum_{i=1}^N \gamma_i g_2^i(\alpha, b, \xi),$$

where

$$f(\alpha, b, \xi) = \frac{1}{2} \sum_{i=1}^N \alpha_i^2 + \sum_{i=1}^N C_i \xi_i + \sum_{i=1}^N \left(1 - \xi_i - y_i \left(\sum_{j=1}^N \alpha_j y_j \mathbf{x}_j \cdot \mathbf{x}_i + \beta_i \right) \right) + \sum_{i=1}^N \beta_i (-\alpha_i) + \sum_{i=1}^N \gamma_i (-\xi_i).$$

2 We can verify that the problem satisfies the condition of the strong duality theorem. Hence, the duality gap is zero.

3 Take partial derivatives of the Lagrangian function L over α, b, ξ yields

$$\frac{\partial}{\partial \alpha_i} L = \alpha_i - \sum_{j=1}^N \omega_j y_j y_i \mathbf{x}_j \cdot \mathbf{x}_i - \beta_i, \quad \frac{\partial}{\partial b} L = - \sum_{i=1}^N \omega_i y_i, \quad \frac{\partial}{\partial \xi_i} L = C_i - \omega_i - \gamma_i.$$

If the following conditions hold,

$$\sum_{i=1}^N \omega_i y_i = 0, \quad \omega_i + \gamma_i = C_i, \quad \forall i = 1, \dots, N,$$

then $\theta(\omega, \beta, \gamma) = L(\alpha, b, \xi, \omega, \beta, \gamma)$ if and only if

$$\alpha_i = \sum_{j=1}^N \omega_j y_j y_i \mathbf{x}_j \cdot \mathbf{x}_i + \beta_i, \quad \forall i \in \{1, N\},$$

at which

$$\theta(\omega, \beta, \gamma) = \sum_{i=1}^N \omega_i - \frac{1}{2} \sum_{i=1}^N \left(\sum_{j=1}^N \omega_j y_j y_i \mathbf{x}_j \cdot \mathbf{x}_i + \beta_i \right)^2.$$

Otherwise, since $\frac{\partial}{\partial \alpha} L, \frac{\partial}{\partial \xi_1} L, \dots, \frac{\partial}{\partial \xi_N} L$ are constants not identically zero, one has $\theta(\omega, \beta, \gamma) = -\infty$.

4 Because $\beta_i \geq 0$,

$$\left(\sum_{j=1}^N \omega_j y_j y_i \mathbf{x}_j \cdot \mathbf{x}_i + \beta_i \right)^2 = \left(\max \left(\sum_{j=1}^N \omega_j y_j y_i \mathbf{x}_j \cdot \mathbf{x}_i, 0 \right) \right)^2 = \left(\sum_{j=1}^N \omega_j y_j y_i \mathbf{x}_j \cdot \mathbf{x}_i \right)_+^2.$$

By (c), one may rewrite (17) as

$$\text{maximize} \quad \sum_{i=1}^N \omega_i - \frac{1}{2} \sum_{i=1}^N \left(\sum_{j=1}^N \omega_j y_j y_i \mathbf{x}_j \cdot \mathbf{x}_i \right)_+^2,$$

subject to

$$\sum_{i=1}^N \omega_i y_i = 0, \quad \omega_i + \gamma_i = C_i, \quad \forall i = 1, \dots, N,$$

variables $\omega_i \geq 0, \beta_i \geq 0, \gamma_i \geq 0, i = 1, \dots, N$, which can be further simplified as (18).

5 By (b), the duality gap is zero, so the primal and dual optimal solutions satisfy the KKT conditions. Note that in (c) we have shown that the stationary condition $\theta(\omega, \beta, \gamma) = L(\alpha, b, \xi, \omega, \beta, \gamma)$ holds iff (21) and (22) are both satisfied. As such, we may write down the KKT conditions, including Stationary, Primal feasible, Dual feasible, and Complementary conditions, as follows (for all $i = 1, \dots, N$):

$$\begin{aligned}
\text{(S1)} \quad \omega_i + \gamma_i &= C_i & \text{(D1)} \quad \omega_i &\geq 0 \\
\text{(S2)} \quad \sum_{i=1}^N \omega_i y_i &= 0 & \text{(D2)} \quad \beta_i &\geq 0 \\
\text{(S3)} \quad \alpha_i &= \sum_{j=1}^N \omega_j y_j y_i \mathbf{x}_j \cdot \mathbf{x}_i + \beta_i & \text{(D3)} \quad \gamma_i &\geq 0 \\
\text{(P1)} \quad y_i(\mathbf{w} \cdot \mathbf{x}_i + b) &\geq 1 - \xi_i & \text{(C1)} \quad \omega_i(1 - \xi_i - y_i(\mathbf{w} \cdot \mathbf{x}_i + b)) &= 0 \\
\text{(P2)} \quad \xi_i &\geq 0 & \text{(C2)} \quad \beta_i \xi_i &= 0 \\
\text{(P3)} \quad \xi_i &\geq 0 & \text{(C3)} \quad \gamma_i \xi_i &= 0
\end{aligned}$$

Given the optimal ω_i , it is clear by (25) and (S3) that $\beta_i = \left(\sum_{j=1}^N \omega_j y_j y_i \mathbf{x}_j \cdot \mathbf{x}_i \right)$ and (19) holds. Given the optimal coefficient vector α , observe that (16) can be rewritten as the following optimization problem:

$$\text{minimize } \frac{1}{2} \sum_{i=1}^N \alpha_i^2 + \sum_{i=1}^N C_i \max \left(1 - y_i \left(\sum_{j=1}^N \alpha_j y_j \mathbf{x}_j \cdot \mathbf{x}_i + b \right), 0 \right),$$

variables $b \in \mathbb{R}, \alpha_i \geq 0, i \in \{1, N\}$, hence the optimal bias is given by (20).

- If $y_i(\mathbf{w} \cdot \mathbf{x}_i + b) < 1$, then $\xi_i > 0$ by (P1), $\gamma_i = 0$ by (C3), $\omega_i = C_i$ by (S1), $\xi_i = 1 - y_i(\mathbf{w} \cdot \mathbf{x}_i + b)$.
- If $y_i(\mathbf{w} \cdot \mathbf{x}_i + b) > 1$, then $\omega_i = 0$ by (P3, C1), so $\gamma_i = C_i$, so $\xi_i = 0$ by (C3).
- If $y_i(\mathbf{w} \cdot \mathbf{x}_i + b) = 1$, then $\xi_i = 0$ by (S1, C1, C2), and $\omega_i = C_i$ by (D1, D3, S1).

Problem 5 (Policy Improvement Theorem for Stochastic Policies)

Consider a discounted Markov Decision Process (MDP) with discount factor $0 \leq \gamma < 1$. A stochastic policy $\pi(a | s)$ induces the Bellman operator $\mathbb{B} \in \ell^\infty(S) \rightarrow \ell^\infty(S)$.

$$(\mathbb{B}^\pi V)(s) = \sum_a \pi(a | s) \left[r(s, a) + \gamma \sum_{s'} \mathbb{P}(s' | s, a) V(s') \right].$$

For any policy π , let V^π and Q^π denote its value function and Q -function.

- (a) Derive the Bellman equations for V^π and Q^π under a stochastic policy:

$$V^\pi(s) = \sum_a \pi(a | s) \left[r(s, a) + \gamma \sum_{s'} \mathbb{P}(s' | s, a) V^\pi(s') \right],$$

$$Q^\pi(s, a) = r(s, a) + \gamma \sum_{s'} \mathbb{P}(s' | s, a) V^\pi(s').$$

Show that V^π is a fixed point of the Bellman operator:

$$V^\pi = \mathbb{B}^\pi V^\pi.$$

- (b) Prove that $\mathbb{B}^\pi$ is a γ -contraction in the ℓ_∞ norm:

$$\|\mathbb{B}^\pi V - \mathbb{B}^\pi W\|_\infty \leq \gamma \|V - W\|_\infty, \quad \forall V, W : \mathcal{S} \rightarrow \mathbb{R}.$$

- (c) Show that $\mathbb{B}^\pi$ has a unique fixed point, and derive its fixed point.

- (d) Let π' be another stochastic policy. Assume that for every state s ,

$$\sum_a \pi'(a | s) Q^\pi(s, a) \geq V^\pi(s).$$

Show that the new policy π' is no worse than π :

$$V^{\pi'}(s) \geq V^\pi(s), \quad \forall s \in \mathcal{S}.$$

(Hint: show that $\mathbb{B}^{\pi'} V^\pi \geq V^\pi$ and show that if $V \leq V'$ then $\mathbb{B}^{\pi'} V \leq \mathbb{B}^{\pi'} V'$)

- (e) Define the stochastic greedy policy π_{greedy} by assigning nonzero probability only to actions

$$\arg \max_a Q^\pi(s, a).$$

Show that π_{greedy} satisfies the condition in part (d), and conclude that

$$V^{\pi_{\text{greedy}}}(s) \geq V^\pi(s), \quad \forall s.$$

(Notice that $S = \arg \max_a Q^\pi(s, a) \subset \mathcal{A}$ may not be a singleton.)

Solution:

- (a) The Bellman operator for the value function is:

$$(\mathbb{B}^\pi V)(s) = \sum_a \pi(a | s) \left[r(s, a) + \gamma \sum_{s'} \mathbb{P}(s' | s, a) V(s') \right].$$

Thus, the Bellman equation for the value function $V^\pi(s)$ is:

$$V^\pi(s) = \sum_a \pi(a | s) \left[r(s, a) + \gamma \sum_{s'} \mathbb{P}(s' | s, a) V^\pi(s') \right].$$

For the $Q^\pi(s, a)$ -function, we have:

$$Q^\pi(s, a) = r(s, a) + \gamma \sum_{s'} \mathbb{P}(s'|s, a) V^\pi(s').$$

We want to show that V^π is a fixed point of the Bellman operator $\mathbb{B}^\pi$, i.e.,

$$V^\pi = \mathbb{B}^\pi V^\pi.$$

Substituting V^π into the Bellman equation for V^π , we get:

$$V^\pi(s) = \sum_a \pi(a|s) \left[r(s, a) + \gamma \sum_{s'} \mathbb{P}(s'|s, a) V^\pi(s') \right],$$

which shows that V^π is indeed a fixed point of the Bellman operator $\mathbb{B}^\pi$.

(b) We need to prove that for any two functions V and W , we have:

$$\|\mathbb{B}^\pi V - \mathbb{B}^\pi W\|_\infty \leq \gamma \|V - W\|_\infty.$$

First, consider:

$$\|\mathbb{B}^\pi V - \mathbb{B}^\pi W\|_\infty = \sup_{s \in \mathcal{S}} \left| \sum_a \pi(a|s) \left[r(s, a) + \gamma \sum_{s'} \mathbb{P}(s'|s, a) V(s') - \left(r(s, a) + \gamma \sum_{s'} \mathbb{P}(s'|s, a) W(s') \right) \right] \right|.$$

Simplifying:

$$\begin{aligned} &= \sup_{s \in \mathcal{S}} \left| \sum_a \pi(a|s) \gamma \sum_{s'} \mathbb{P}(s'|s, a) [V(s') - W(s')] \right| \\ &\leq \gamma \sup_{s \in \mathcal{S}} \sum_a \pi(a|s) \sum_{s'} \mathbb{P}(s'|s, a) |V(s') - W(s')|. \end{aligned}$$

Since $|V(s') - W(s')| \leq \|V - W\|_\infty$ for all s' , we get:

$$\|\mathbb{B}^\pi V - \mathbb{B}^\pi W\|_\infty \leq \gamma \|V - W\|_\infty \sup_{s \in \mathcal{S}} \sum_a \pi(a|s) \sum_{s'} \mathbb{P}(s'|s, a) = \gamma \|V - W\|_\infty.$$

Thus, $\mathbb{B}^\pi$ is a γ -contraction in the ℓ_∞ norm.

(c) Since $\mathbb{B}^\pi$ is a contraction mapping in the ℓ_∞ norm with contraction constant $\gamma < 1$, by the Banach Fixed Point Theorem, $\mathbb{B}^\pi$ has a unique fixed point V^π . This fixed point satisfies:

$$V^\pi = \mathbb{B}^\pi V^\pi.$$

To use operator to describe the $V^\pi = (\mathbf{I} - \gamma \mathbb{P})^{-1} r$

(d)

$$\sum_a \pi'(a|s) Q^\pi(s, a) \geq V^\pi(s).$$

We want to show that $\mathbb{B}^{\pi'} V^\pi(s) \geq V^\pi(s)$ for all $s \in \mathcal{S}$.

First,

$$\mathbb{B}^{\pi'} V^\pi(s) = \sum_a \pi'(a|s) (r(s, a) + \gamma \sum_{s'} \mathbb{P}(s'|s, a) V^\pi(s')) = \sum_a \pi'(a|s) Q^\pi(s, a) \geq V^\pi(s)$$

Second, If $V \leq V'$ then $\mathbb{B}^{\pi'} V \leq \mathbb{B}^{\pi'} V'$

$$\mathbb{B}^{\pi'} V(s) = \sum_a \pi'(a|s) (r(s, a) + \gamma \sum_{s'} \mathbb{P}(s'|s, a) V(s')) \leq \sum_a \pi'(a|s) (r(s, a) + \gamma \sum_{s'} \mathbb{P}(s'|s, a) V'(s')) = \mathbb{B}^{\pi'} V'$$

Therefore, $V^\pi(s) \leq \mathbb{B} V^\pi(s) \leq \mathbb{B}^2 V^\pi(s) \leq \dots \leq V^{\pi'}(s) \forall s \in \mathcal{S}$

(e)

$$\pi_{\text{greedy}}(a|s) = \begin{cases} p_a \neq 0, & \text{if } a \in \arg \max_a Q^\pi(s, a), \\ 0, & \text{otherwise.} \end{cases}$$

Furthermore, we have

$$Q^\pi(s, a_{\text{greedy}}) = \max_a Q^\pi(s, a), \forall a_{\text{greedy}} \in \arg \max_a Q^\pi(s, a)$$

We may observe the condition for (d) is that

$$\begin{aligned} V^\pi(s) &= \sum_a \pi(s, a) Q^\pi(s, a) \leq \max_a Q^\pi(s, a) \sum_a \pi(s, a) = \max_a Q^\pi(s, a) = \sum_a \pi_{\text{greedy}}(s, a) \max_a Q^\pi(s, a) \\ &= \sum_a \pi_{\text{greedy}}(s, a) Q^{\pi_{\text{greedy}}}(s, a) = V^{\pi_{\text{greedy}}}(s) \end{aligned}$$